$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{4\tan^2 x + 5}\, dx$$ {(Dividing both numerator and denominator by $\cos^2 x$).

Let $u = \tan x$. Then $du = \sec^2 x\, dx$

When $x = 0, u = \tan 0 = 0$

When $x = \frac{\pi}{2}, u = \tan\frac{\pi}{2} = \infty$.

$\therefore I = \int_0^{\infty} \frac{du}{4u^2 + 5}$ (This is an improper integral)

$$= \frac{1}{4}\int_0^{\infty} \frac{du}{\left[u^2 + \left(\frac{\sqrt{5}}{2}\right)\right]} = \frac{1}{4} \times \frac{2}{\sqrt{5}} \left[\tan^{-1}\left(\frac{u}{\frac{\sqrt{5}}{2}}\right)\right]_0^{\infty} = \frac{1}{2\sqrt{5}}(\tan^{-1}\infty - \tan^{-1} 0) = \frac{1}{2\sqrt{5}}\left(\frac{\pi}{2}\right) = \frac{\pi}{4\sqrt{5}}.$$

EXERCISE 9.5

1. Evaluate the following:

(i) $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + 5\cos^2 x}$ (ii) $\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 4\sin^2 x}$

9.6 Reduction Formulae

Certain definite integrals can be evaluated by an index-reduction method. In this section, we obtain the values of the following definite integrals:

$\int_0^{\frac{\pi}{2}} \sin^n x\, dx$, $\int_0^{\frac{\pi}{2}} \cos^n x\, dx$, $\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x\, dx$, $\int_0^1 x^m (1-x)^n dx$.

We also obtain the value of the improper integral $\int_0^{\infty} e^{-x} x^n\, dx$.

The method of obtaining a reduction formula has the following steps:

Step 1 : Identify an index (positive integer) n in the integral.

Step 2 : Put the integral as I_n.

Step 3 : Applying integration by parts, obtain the equation for I_n in terms of I_{n-1} or I_{n-2}.

The resulting equation is called the reduction formula for I_n.

We list below a few reduction formulae without proof:

Reduction Formula I : If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, then $I_n = \frac{(n-1)}{n} I_{n-2}, n \geq 2$.

Reduction Formula II : If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$, then $I_n = \frac{(n-1)}{n} I_{n-2}\ n \geq 2$.

Reduction Formula III : If $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x\, dx$, then $I_{m,n} = \frac{(n-1)}{m+n} I_{m,n-2}, n \geq 2$.

Reduction Formula IV : If $I_{m,n} = \int_0^1 x^m (1-x)^n dx$, then $I_{m,n} = \frac{n}{m+n+1} I_{m,n-1}, n \geq 1$.

Chapter 9 Applications of Integration.indd 117 17-02-2020 17:06:04